

Real-Time Public Sentiment Dashboard Using Twitter Data

By: Sherin Samuel

Date: 03/05/2025

Abstract

This project presents a Real-Time Public Sentiment Dashboard designed to monitor public opinion about Tesla using Twitter data. Due to limitations in API access, manual tweet collection was used to simulate real-time monitoring. Sentiment analysis was performed using Python-based tools, and the results were visualized using Python libraries. Although a Tableau dashboard was initially planned, it was not deployed in this version due to API constraints.

Introduction

Twitter provides a rich source of real-time public opinion that can be analyzed to understand brand perception. This project aimed to analyze sentiment in Tesla-related tweets using natural language processing (NLP) techniques and to visualize sentiment trends over time. Despite API limitations, the project simulates a working pipeline and provides meaningful sentiment insights.

Tools Used

- **Programming:** Python
- **Libraries:** Pandas, NLTK, TextBlob, Matplotlib, Tenacity
- **Data Storage:** CSV, Google Sheets API
- **Development Environment:** VS Code, Jupyter Notebook

Steps Involved in Building the Project

1. **Data Collection:**
 - Tesla-related tweets were manually collected (due to API issues).
 - Tweets were stored in CSV format with timestamps.
2. **Preprocessing:**
 - Cleaned tweets by removing URLs, mentions, hashtags, and punctuation.
 - Text normalized to lowercase, tokenized, and stopwords removed.
3. **Sentiment Analysis:**

- TextBlob used to calculate sentiment polarity.
 - Classification rules:
 - **Positive:** Polarity > 0
 - **Negative:** Polarity < 0
 - **Neutral:** Polarity = 0
 - 4. **Aggregation:**
 - Sentiment counts summarized daily.
 - Output stored in daily_summary.csv.
 - 5. **Visualization:**
 - Python (Matplotlib) used to create sentiment trend charts.
 - Tableau dashboard development was planned but not executed due to API constraints.
-

Conclusion

This project effectively simulates real-time sentiment tracking using manual tweet data and Python analytics. The pipeline is ready for full automation once Twitter API access is available. Future plans include real-time streaming integration, Tableau dashboard deployment, and support for multilingual sentiment analysis.